Abenezer Girma **Taye**

Postdoctoral Research Scholar

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SUMMARY

I am a researcher and educator specializing in safe, secure, and sustainable autonomy for advanced aerial mobility (AAM) applications. I develop **decision-making, planning, and control algorithms** for autonomous systems operating in uncertain, energy-constrained, and adversarial environments. Currently, I am a postdoctoral research scholar at the Institute for Software Integrated Systems at Vanderbilt University. Prior to Vanderbilt, I received my Ph.D. in Aerospace Engineering at The George Washington University, and completed two research internships at MathWorks.

RESEARCH INTERESTS

Theory: Optimal Control, Reinforcement Learning, Physics-Informed Machine Learning, Optimization, Estimation Theory.
Application: Multi-agent Trajectory Planning, Air Traffic Management, Aviation Electrification, Risk Assessment and Contingency Management, Security of Cyber-Physical Systems.

Testing Platforms for Novel AAM Concepts
• eVTOL Simulators • sUAS Hardware Flight Testing • Battery Digital Twins • UAM/UAS Transportation Networks

EDUCATION

George Washington University — *Washington, DC*
Ph.D. in Aerospace Engineering, — *Jan. 2021 - Aug. 2025*
Dissertation title: *Enhancing Safety and Energy Efficiency in Advanced Air Mobility Through Trajectory Planning and Mission Feasibility Assessment Strategies*
Advisor: Prof. Peng Wei

North Carolina Agricultural and Technical State University — *Greensboro, North Carolina*
Ph.D. in Electrical and Computer Engineering (Transferred Before Completion) — *Aug. 2019 - Dec. 2020*

Addis Ababa Science and Technology University — *Addis Ababa, Ethiopia*
B.S. in Electro-Mechanical Engineering, — *Sep. 2012 - June 2017*
Thesis: *Modeling and Control of an Autonomous Forklift for EPHARM Pharmaceuticals*
Advisor: Prof. Solomon Gudeta
Class Rank: 1st out of 183, GPA: 3.97/4.00

RESEARCH & INDUSTRY EXPERIENCE

Postdoctoral Research Scholar, Vanderbilt University — *Nashville, Tennessee*
Supervisors: Prof. Bryan C. Ward, Prof. Gautam Biswas — *Oct. 2025 - present*
Focus: Securing AAM operations and cyber-physical systems against cyberattacks

Research Intern, Sensor Fusion and Tracking Toolbox, MathWorks — *Natick, Massachusetts*
Supervisors: Prashant Arora, Dr. Elad Kivelevitch — *May 2023 - Aug. 2023*

– Conducted research on multi-object tracking algorithms (JPDA, PHD, MHT) to facilitate in-depth performance evaluation
– Built a MATLAB application to design and analyze multi-object tracking algorithms
– Demonstrated effectiveness using real-world radar data in aircraft landing approach scenarios

Research Intern, MathWorks Advanced Research and Technology Office, MathWorks — *Natick, Massachusetts*
Supervisors: Dr. Roberto Valenti, Dr. Pieter Mosterman — *May 2022 - Aug. 2022*

– Developed a decentralized, highly scalable trajectory planning framework for urban air mobility
– Developed an online learning algorithm to approximate aircraft reachable sets via a discrepancy function
– Demonstrated a significant reduction in near-midair collisions in large-scale simulations

MAJOR PROJECTS

Securing High-Density Urban Airspaces *Oct. 2025 - Onward*
Supervisors: Prof. Bryan C. Ward, Prof. Gautam Biswas

- Leading a team of six research engineers on a **$6M NASA-funded University Leadership Initiative (ULI)** project
- Developing algorithms to identify vulnerabilities in AAM systems
- Designing defense strategies against cyberattacks and improving system-level resilience and security

In-Time Learning-Based Safety Management for Scalable AAM Operations *Mar. 2021 - May 2024*
Supervisor: Prof. Peng Wei

- Led the first thrust of a **$2.5M NASA-funded System-Wide Safety** project
- Proposed a mission risk assessment framework to evaluate the impact of adverse weather on flight safety
- Demonstrated the framework's effectiveness in real-world-inspired scenarios, including drone package delivery and urban air taxi services

Safe, Scalable, and Energy-Efficient Trajectory Planner for AAM Operations *Jan. 2021 - Aug. 2025*
Supervisor: Prof. Peng Wei

- Developed a real-time, multi-agent trajectory planner for AAM operations
- Utilized a data-driven reachability analysis approach for on-the-fly safety verification
- Proposed a scalable Markov decision process (MDP) framework to optimize trajectories for energy efficiency

Learning-based Li-ion Battery Modeling for Contingency Management *Sept. 2022 - Aug. 2025*
Supervisor: Prof. Peng Wei

- Developed a data-driven battery model using Neural ODEs and Physics-Informed Neural Networks to address the computational limitations of traditional electrochemical models
- Achieved a 33× speedup in battery state prediction and mission risk assessment compared to model-based approaches
- Illustrated the efficacy of the method on a realistic drone package delivery scenario

Energy Demand Analysis for eVTOL Aircraft in Urban Air Mobility *Sept. 2023 - Dec. 2024*
Supervisor: Prof. Peng Wei

- Developed a predictive framework to forecast charging demand for electric vertical takeoff and landing (eVTOL) aircraft
- Integrated energy consumption models with mission specifications, wind forecasts, and charging behaviors to estimate aggregated power demand at vertiports.
- Illustrated the efficacy of the framework on a large scale realistic air taxi operation

TEACHING EXPERIENCE

Academy of College Teaching (ACT) Certificate (Description)
Awarded by Vanderbilt University Institute for the Advancement of Higher Education (AdvancED) *Spring 2026*

Graduate Teaching Assistantship Program (GTAP) Certificate (Description)
Program organized by GWU for graduate students to master teaching skills *Fall 2024*

Teaching Assistant
GWU MAE 4182 - Electro-Mechanical Control Systems *Fall 2024*
Role: Substitute lectures, weekly office hours and recitations, creating assignments

GWU Informal Seminar - Verifying Cyber-Physical Systems *2022 - 2023*
Role: Initiating the seminar, presenting each chapter from the book *Verifying Cyber-Physical Systems*, leading discussions

NCA&T ECEN 4682/6682 - Genetic Algorithm *Spring 2020*
Role: Weekly office hours and recitations, mentorship on final projects, mentorship on academic and professional life

MENTORSHIP EXPERIENCE

Undergraduate Students
Jairo Sanchez, Mechanical and Aerospace Engineering, George Washington University *Spring 2023*
Tharun Saravanan, Computer Science, George Washington University *Summer 2023*
Liza Mozolyuk, Mechanical and Aerospace Engineering, George Washington University *Summer 2023*
Jeremiah Webb, Computer Science, Vanderbilt University *(Resulted in 1 DASC publication)* *Fall 2025 - present*

Graduate Students
Alex Zango, Mechanical and Aerospace Engineering, George Washington University *Spring 2025*
Bryce Bjorkman, Computer Science, Vanderbilt University *Fall 2025 - present*

TECHNICAL SKILLS

Programming Skills:	Python, MATLAB/Simulink, C/C++, Julia
Packages:	PyTorch, PSOPT, Gurobi, BlueSky, Gymnasium
Operating Systems:	Linux, Windows, ROS
Standards:	Familiar with DO-178C certification process

PUBLICATIONS

Journal Publications

Published, in press, submitted

1. **A. Taye**, P. Wei, "Energy-Efficient Trajectory Planning and Mission Feasibility Assessment Framework for Drone Package Delivery Operations", *AIAA Journal of Aerospace Information Systems*, vol. 22, no. 7, July 2025.
2. **A. Taye**, R. Valenti, A. Rajhans, A. Mavrommati, P.J. Mosterman and P. Wei, "Safe and Scalable Real-Time Trajectory Planning Framework for Urban Air Mobility", *AIAA Journal of Aerospace Information Systems*, vol. 21, no. 8, Aug. 2024.
3. P. Razzaghi, A. Tabrizian, W. Guo, S. Chen, **A. Taye**, E. Thompson, A. Bregeon, A. Baheri and P. Wei, "A Survey on Reinforcement Learning in Aviation Applications", *Engineering Applications of Artificial Intelligence,* vol. 136, part A, Oct. 2024.

Conference Publications

Published, in press, accepted

1. J. Webb, B. Bjorkman, A. Diaz Gonzalez, A. Coursey, N. Dahle, K. Lemieux Mack, F. Fotiadis, G. Biswas, B. C. Ward, **A. Taye,** "Remote ID Spoofing-Aware Trajectory Planning for Small Unmanned Aerial Systems," to appear in AIAA/IEEE Digital Avionics Systems Conference, Orlando, FL, 2026.
2. N. Dahle, R. Canady, A. Diaz-Gonzalez, A. Coursey, B. Bjorkman, C. Lemieux-Mack, S. Gonzalez, N. Koroma, **A. Taye**, G. Biswas, B. C. Ward, X. Koutsoukos "Detection of Compromised UAVs using Graph Machine Learning," AIAA SciTech Forum, Orlando, FL, 2026.
3. A. Diaz-Gonzalez, A. Coursey, B. Bjorkman, D. Shatokhin, C. Lemieux-Mack, S. Gonzalez, N. Dahle, N. Koroma, **A. Taye**, R. Canady, X. Koutsoukos, G. Biswas, B. C. Ward "Networked Simulation for Cybersecurity Evaluation of Small Unmanned Aircraft Systems in Dense Urban Environments," AIAA SciTech Forum, Orlando, FL, 2026.
4. B. Bjorkman, S. Zheng, A. Coursey, C. Lemieux-Mack, S. Gonzalez, A. Diaz-Gonzalez, N. Dahle, N. Koroma, R. Canady, X. Koutsoukos, G. Biswas, **A. Taye**, B. C. Ward "Remote ID Spoofing Attacks and Defenses," AIAA SciTech Forum, Orlando, FL, 2026.
5. I. Sharifi, M. Ghazanfari, **A. Taye**, P. Wei, M. H. Ahmed, H. T. Kim, M. Ghasemi, V. Gupta, N. Dahle, R. Canady, A. D. Gonzalez, A. Coursey, B. Bjorkman, et al. "A Survey of Security Challenges and Solutions for UAS Traffic Management (UTM) and small Unmanned Aerial Systems (sUAS)," AIAA SciTech Forum, Orlando, FL, 2026.
6. M. Ghazanfari, I. Sharifi, **A. Taye**, P. Wei, B. C. Ward, X. Koutsoukos, G. Biswas, N. Dahle, R. Canady, A. D. Gonzalez, A. Coursey, B. Bjorkman, et al. "A Survey of Security Challenges and Solutions for Advanced Air Mobility and eVTOL Aircraft," AIAA SciTech Forum, Orlando, FL, 2026.
7. **A. Taye**, A. Coursey, M. Quinones-Grueiro, C. Hu, G. Biswas, P. Wei, "Safe to Fly? Real-Time Flight Mission Feasibility Assessment for Drone Package Delivery Operations", The 36th International Conference on Principles of Diagnosis and Resilient Systems (DX'25), Nashville, TN, Sept. 2025.
8. C. Vu, M. Ghazanfari, K Dong, **A. Taye**, A. Tabrizian, and P. Wei "Transformer or CNN? Benchmarking Real-Time Detection Transformer and YOLOv8 for Small UAS Autonomous Landing", AIAA AVIATION, Las Vegas, NV, July 2025.
9. J. Xu, C. Vu, **A. Taye**, A. Tabrizian, P. Wei, "Small UAS Landing Site Detection with ArUco Markers and Deep Learning based Computer Vision", AIAA AVIATION, Las Vegas, NV, July 2025.
10. D. Ding, C. Vu, **A. Taye**, A. Tabrizian, P. Wei, Z. Zhao, "Synthetic Data Generation for Computer Vision based Autonomous Landing for Small UAS Package Delivery", AIAA AVIATION, Las Vegas, NV, July 2025.
11. **A. Taye**, S. Chen, P. Wei, "Energy-Aware Strategic Traffic Management for Urban Air Mobility", AIAA SCITECH, Orlando, FL, Jan 2025.
12. A. Tabrizian, P. Gupta, **A. Taye**, J. Jones, E. Thompson, S. Chen, T. Bonin, D. Eberle and P. Wei, "Using Large Language Models to Automate Flight Planning under Wind Hazards", AIAA/IEEE Digital Avionics Systems Conference (DASC), San Diego, CA, Sept. 2024.

13. **A. Taye**, P. Wei, P. Pradeep, J. Jones, T. Bonin, and D. Eberle, "Energy Demand Analysis for eVTOL Charging Stations in Urban Air Mobility", AIAA AVIATION, Las Vegas, NV, July 2024.
14. **A. Taye**, P. Wei, "Flight Mission Feasibility Assessment of Urban Air Mobility Operations under Battery Energy Constraint", AIAA SCITECH, Orlando, FL, Jan. 2024.
15. **A. Taye**, E. Thompson, P. Wei, T. Bonin, and J. Jones, "Probabilistic Evaluation for Flight Mission Feasibility of a Small Octocopter in the Presence of Wind", AIAA AVIATION, San Diego, CA, Jun. 2023.
16. **A. Taye**, J. Bertram, C. Fan, and P. Wei, "Reachability-based Online Safety Verification for High-density Urban Air Mobility Trajectory Planning", AIAA AVIATION, Chicago, IL, Jun. 2022.
17. E. Thompson, **A. Taye**, W. Guo, P. Wei, M. Quinones, I. Ahmed, G. Biswas, J. Quattrociocchi, S. Carr, U. Topcu, and J. Jones, "A Survey of eVTOL Aircraft and AAM Operation Hazards." AIAA AVIATION, Chicago, IL, Jun. 2022.

Technical Reports

1. **A. Taye** and P. Wei, "AI Safety and Certification in Aviation Systems - Identifying the Gaps and Opportunities", Aerospace Industries Association (AIA), 2022.

INVITED TALKS

1. **A. Taye** "Safe to Fly? Real-Time Flight Mission Feasibility Assessment for Drone Package Delivery Operation", The 36th International Conference on Principles of Diagnosis and Resilient Systems (DX'25), Nashville, TN, Sept. 2025.
2. **A. Taye** "Energy-Aware Strategic Traffic Management for Urban Air Mobility", AIAA SCITECH, Orlando, FL, Jan 2025.
3. **A. Taye** "Energy-Efficient Trajectory Planning and Mission Feasibility Assessment Framework for Drone Package Delivery Operations", INFORMS Annual Meeting, Seattle, WA, Oct. 2024.
4. **A. Taye** "Energy Demand Analysis for eVTOL Charging Stations in Urban Air Mobility", AIAA AVIATION, Las Vegas, NV, July 2024.
5. **A. Taye** "Flight Mission Feasibility Assessment of Urban Air Mobility Operations under Battery Energy Constraint", AIAA SCITECH, Orlando, FL, Jan. 2024.
6. **A. Taye**, J. Bertram, C. Fan, and P. Wei, "Reachability based online safety verification for high-density urban air mobility trajectory planning", AIAA AVIATION, Chicago, IL, Jun. 2022.

HONORS AND AWARDS

Honorable Mention Award, AIAA Young Professionals, Students, and Educators (YPSE) Conference	*2022*
Student Travel Grant for AIAA AVIATION	*2022, 2024*
Student Travel Grant for AIAA SCITECH	*2024, 2025*
Ambassador of Science and Technology of the Graduation Class of 2017, Ministry of Science and Technology of Ethiopia	*2017*
Mid Academic Life Prize, AASTU	*2014*

PROFESSIONAL AFFILIATIONS

American Institute of Aeronautics and Astronautics (AIAA), Member
Institute of Electrical and Electronic Engineers (IEEE), Member
Institute for Operations Research and the Management Sciences (INFORMS), Member

PROFESSIONAL SERVICE

Conference Session and Workshop Organizer

Workshop Co-Host *Apr. 2025*
"NASA System-Wide Safety Program Workshop"
George Washington University, Washington DC

Session Co-Chair *Aug. 2024*
"System-Wide Safety"
AIAA Aviation 2024, Las Vegas, NV

Session Co-Chair *Aug. 2024*
"Regional Air Mobility and General Aviation"
AIAA Aviation 2024, Las Vegas, NV

Reviewer Activities

- *Journal referee*

- Aerospace Science and Technology *2022 - 2026*
- The Aeronautical Journal *2023 - 2026*
- AIAA Journal of Aerospace Information Systems *2022 - 2026*
- IEEE Transactions on Intelligent Transportation Systems *2022 - 2026*

- *Conference referee*
  - AIAA Aviation *2022 - 2026*
  - AIAA SciTech *2022 - 2026*

REFERENCES

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